ClipSAT : Contest Math & Olympiad AMC 10/12, Kangaroo & Mawhiba [ANSWER KEY]

Unit 1.01: Olympiad Algebraic Manipulation, Vieta & Inequalities

Q1. The coordinate diagram shows the parabolic curve f_1x = x^2 - 6x + 20 in the domain of Olympiad Algebra, Vieta & Inequalities. By completing the square f_1x = x - h^2 + k , find the sum k + f_15 .

[x] A: 26

[ ] B: 29

[ ] C: 15

[ ] D: 24

Solution: **Mark Scheme & Examiner Solution Olympiad Algebra, Vieta & Inequalities:** 1. Complete the square: x^2 - 6x + 20 = x - 3^2 - 9 + 20 = x - 3^2 + 11 , giving minimum value k = 11 . **[M1]** 2. Evaluate at x = 5 : f_15 = 2^2 + 11 = 15 . **[M1]** 3. Sum the two quantities: k + f_15 = 11 + 15 = 26 . **[A1]** - **Final Verified Answer:** 26 .

Q2. A mathematical model in Olympiad Algebra, Vieta & Inequalities is defined by the cubic function y_2x = 2x^3 - 3x^2 + 15x - 8 . Calculate the exact gradient of the tangent to the curve at the point where x = 3 .

[ ] A: 69

[x] B: 51

[ ] C: 36

[ ] D: 27

Solution: **Mark Scheme & Examiner Solution Olympiad Algebra, Vieta & Inequalities:** 1. Differentiate term by term using the power rule: frac{mathrm{d}y}{mathrm{d}x} = 6x^2 - 6x + 15 . **[M1]** 2. Substitute x = 3 : left.frac{mathrm{d}y}{mathrm{d}x}right|_{x=3} = 69 - 63 + 15 = 51 . **[M1][A1]** - **Final Verified Answer:** 51 .

Q3. In Olympiad Algebra, Vieta & Inequalities, two displacement vectors are given by mathbf{u}_{3} = langle 3, -8 rangle and mathbf{v}_{3} = langle 9, 7 rangle . Compute the exact scalar value of mathbf{u}_{3} cdot mathbf{v}_{3} + |mathbf{u}_{3}|^2 .

[ ] A: -29

[ ] B: 73

[x] C: 44

[ ] D: 156

Solution: **Mark Scheme & Examiner Solution Olympiad Algebra, Vieta & Inequalities:** 1. Compute the inner product: mathbf{u}_{3} cdot mathbf{v}_{3} = 39 + -87 = 27 + -56 = -29 . **[M1]** 2. Compute the squared norm: |mathbf{u}_{3}|^2 = 3^2 + -8^2 = 73 . **[M1]** 3. Combine: -29 + 73 = 44 . **[A1]** - **Final Verified Answer:** 44 .

Q4. The linear growth profile shown in the diagram for Olympiad Algebra, Vieta & Inequalities follows an arithmetic progression with first term a_1 = 4 and common difference d = 6 . Find the exact sum S_{15} of the first 15 terms.

[ ] A: 88

[ ] B: 780

[ ] C: 1320

[x] D: 690

Solution: **Mark Scheme & Examiner Solution Olympiad Algebra, Vieta & Inequalities:** 1. Use the sum formula for an arithmetic series: S_n = frac{n}{2}[2a_1 + n - 1d] . **[M1]** 2. Substitute n = 15, a_1 = 4, d = 6 : S_{15} = frac{15}{2}[8 + 14 times 6] = frac{15}{2} times 92 = 690 . **[M1][A1]** - **Final Verified Answer:** 690 .

Unit 1.01: Olympiad Algebraic Manipulation, Vieta & Inequalities

Q5. In Olympiad Algebra, Vieta & Inequalities, evaluate the exact definite integral I_5 = int_0^{2} 18x^2 + 12x,mathrm{d}x .

[x] A: 72
[ ] B: 96
[ ] C: 84
[ ] D: 37

Solution: **Mark Scheme & Examiner Solution Olympiad Algebra, Vieta & Inequalities:** 1. Integrate term by term: int 18x^2 + 12x,mathrm{d}x = left[6x^3 + 6x^2right]_0^{2} . **[M1]** 2. Substitute the upper and lower limits: 62^3 + 62^2 - 0 = 48 + 24 = 72 . **[M1][A1]** - **Final Verified Answer:** 72 .

Q6. In Olympiad Algebra, Vieta & Inequalities, consider the quadratic function f_6x = x^2 - 8x + 99 . Express f_6x in the completed-square form x - h^2 + k and compute the exact value of k + f_611 .

[ ] A: 219
[x] B: 215
[ ] C: 132
[ ] D: 208

Solution: **Mark Scheme & Examiner Solution Olympiad Algebra, Vieta & Inequalities:** 1. Complete the square: x^2 - 8x + 99 = x - 4^2 - 16 + 99 = x - 4^2 + 83 , giving minimum value k = 83 . **[M1]** 2. Evaluate at x = 11 : f_611 = 7^2 + 83 = 132 . **[M1]** 3. Sum the two quantities: k + f_611 = 83 + 132 = 215 . **[A1]** - **Final Verified Answer:** 215 .

Q7. The graph illustrates the cubic model y_7x = 2x^3 - 4x^2 + 33x - 15 studied in Olympiad Algebra, Vieta & Inequalities. Find the instantaneous gradient frac{mathrm{d}y}{mathrm{d}x} of the tangent to the curve at x = 8 .

[ ] A: 417
[ ] B: 320
[x] C: 353
[ ] D: 73

Solution: **Mark Scheme & Examiner Solution Olympiad Algebra, Vieta & Inequalities:** 1. Differentiate term by term using the power rule: frac{mathrm{d}y}{mathrm{d}x} = 6x^2 - 8x + 33 . **[M1]** 2. Substitute x = 8 : left.frac{mathrm{d}y}{mathrm{d}x}right|_{x=8} = 664 - 88 + 33 = 353 . **[M1][A1]** - **Final Verified Answer:** 353 .

Q8. In Olympiad Algebra, Vieta & Inequalities, two displacement vectors are given by mathbf{u}_{8} = langle 4, -7 rangle and mathbf{v}_{8} = langle 8, 5 rangle . Compute the exact scalar value of mathbf{u}_{8} cdot mathbf{v}_{8} + |mathbf{u}_{8}|^2 .

[ ] A: -3
[ ] B: 65
[ ] C: 132
[x] D: 62

Solution: **Mark Scheme & Examiner Solution Olympiad Algebra, Vieta & Inequalities:** 1. Compute the inner product: mathbf{u}_{8} cdot mathbf{v}_{8} = 48 + -75 = 32 + -35 = -3 . **[M1]** 2. Compute the squared norm: |mathbf{u}_{8}|^2 = 4^2 + -7^2 = 65 . **[M1]** 3. Combine: -3 + 65 = 62 . **[A1]** - **Final Verified Answer:** 62 .